QuickSpot®

Walter - WiFi/BLE/NB-IoT/LTE-M/GNSS module

Datasheet



1 General information

Walter is a small-form-factor IoT system-on-module (SoM) that combines a powerful ESP32-S3 system-on-chip (SoC) with a second-generation Sequans GM02SP LTE-M/NB-IoT 5G modem and GNSS receiver. Its ESP32-S3 provides many built-in peripherals as well, including UART, SPI, I²C, CAN, Wi-Fi b/g/n, and Bluetooth 5.

2 Features

The default version of Walter is based on an ESP32-S3-WROOM-1-N16R2 module (16MB flash, 2MB PSRAM, PCB antenna) with an on-board Sequans GM02SP modem. Versions with 8MB PSRAM and/or a u.FL connector for an external WiFi/BLE antenna are optional and available through direct ordering (see section 12). This combination makes Walter a unique development board that offers a rich feature-set which include but is not limited to:

- CPU: Xtensa Dual-core 32-bit LX7 CPU (ESP32-S3 SoC)
- RAM: 2MB (Quad SPI) PSRAM, 8MB (Octal SPI) PSRAM optional
- Flash: 16MB (Quad SPI) Flash memory
- WiFi: 150Mbps(n) 802.11 WiFi b/g/n with on-board PCB antenna, u.FL connector optional
- LTE: CAT M1/NB1/NB2 (GM02SP module)
- GPS: GPS, GNSS Constellation support (GM02SP module)
- Bluetooth: 2Mbps Bluetooth 5 (LE), Bluetooth Mesh
- 24 physical GPIO pins
- Ultra low deep sleep current of 9.5µA
- Open source software support for ESP-IDF, Arduino, Micropython and Toit
- Certified for CE, FCC, IC, UKCA, New-Zealand and Australia, and approved by Verizon
- Solderless integration into any project due to the 2.54mm pin headers

3 Electrical characteristics

For the most reliable use and stability of the module we advise to use the typical ratings. We do not guarantee the correct functioning of the device outside the minimum and maximum range of the module.

Parameter	Units	Minimum rating	Typical rating	Maximum rating
Supply voltage VIN, functional	V	3.0	5.0	5.5[3]
Supply voltage VIN, recommended	V	3.3	5.0	5.5[3]
Digital I/O voltage	V	2.64	3.3	3.6
Peak input current @ VIN = 3.0V[1]	A	–	–	1.5
Peak input current @ VIN = 5.0V[1]	A	–	–	1.0
3V3 OUT output current	A	–	–	0.25
3V3 OUT capacitive load	µF	–	–	1
Deep sleep current[2]	µA	–	9.5	–

[1] Worst case with LTE transmitting at 23dBm, WiFi transmitting and 250mA drawn from 3V3 OUT at the same time, including a 20% margin. Use these values to size the power supply. Typical values are given in section 5.1.3.
[2] Measured at VIN = 5.0V with the modem in PSM.
[3] 5.5V is also the absolute maximum rating of VIN. VIN must never exceed 5.5V, not even during transients. See section 5.1.1 for details.

4 Interfaces

Walter provides a total of 28 physical pins (3 power, 1 strapping pin and 24 I/O pins) to interface with external parts. This chapter provides information about these pins as well as internally connected pins and the testpoints located at the bottom of the board.

Power supply pins and their details are available in section 5.1 about the power characteristics.

For more information about specific pins regarding the ESP32-S3 Wroom module or the Sequans GM02SP module, please refer to the datasheet of the corresponding module.

4.1 Pin Assignment

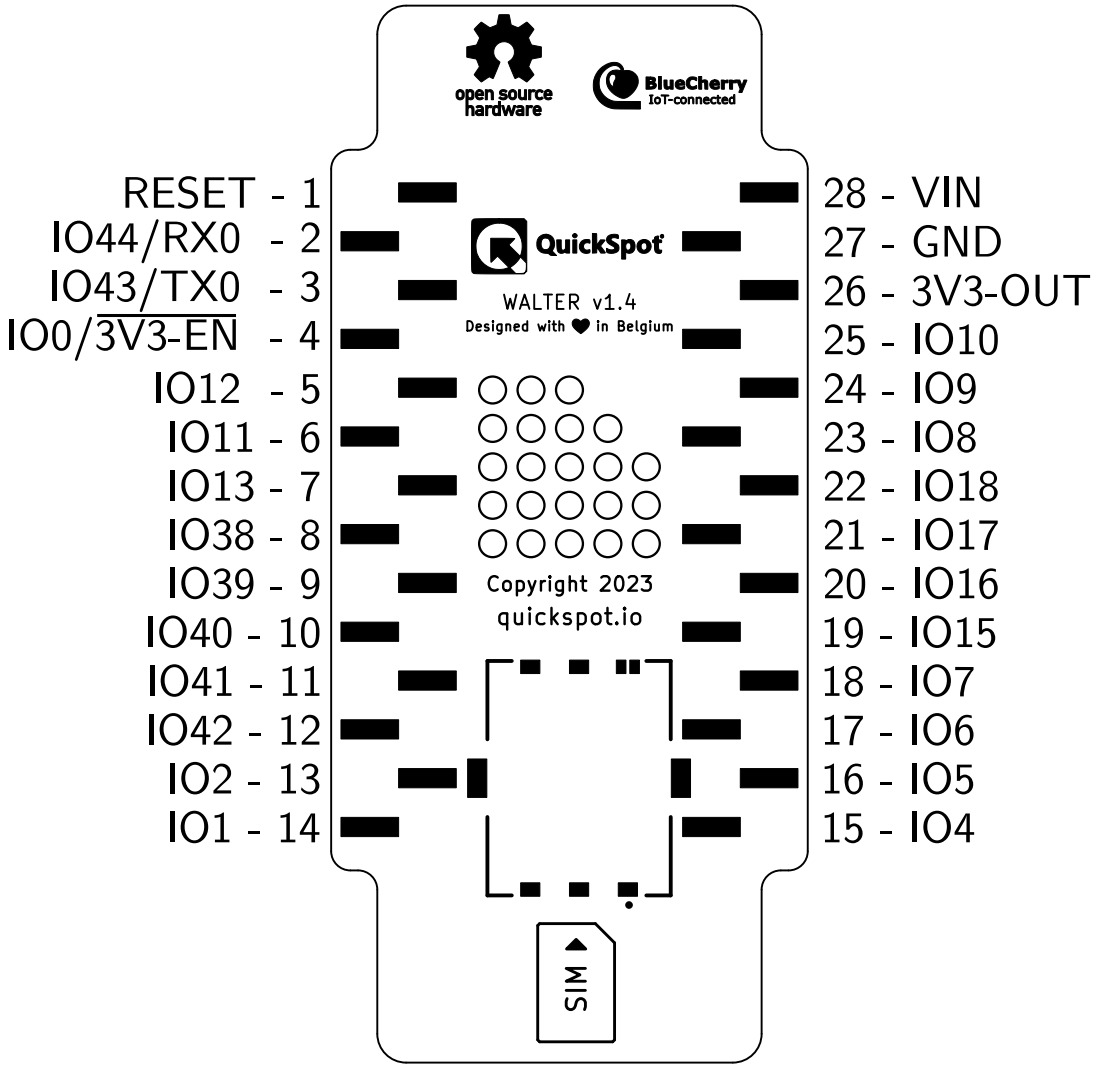


Figure 1: Walter pinout (top view)

4.1.1 External Pins

Table 1 contains the description of the physical pins on Walter available on the underside of the board. The order of this table, in reference to the board, is top to bottom and left to right.

Pin	Function	ESP pin	Input/Output	Description
1	RESET	EN	input	ESP32 reset with 10k pullup
2	IO44/RX0	RXD0	bidirectional	ESP32 UART0 Receive
3	IO43/TX0	TXD0	bidirectional	ESP32 UART0 Transmit
4	$\overline{\text{DFU}}$/$\overline{\text{3V3_EN}}$	IO0	bidirectional	DFU when low on boot, 3V3 OUT on when low
5	IO12	IO12	bidirectional	General purpose I/O
6	IO11	IO11	bidirectional	General purpose I/O
7	IO13	IO13	bidirectional	General purpose I/O
8	IO38	IO38	bidirectional	General purpose I/O
9	IO39	IO39	bidirectional	General purpose I/O
10	IO40	IO40	bidirectional	General purpose I/O
11	IO41	IO41	bidirectional	General purpose I/O
12	IO42	IO42	bidirectional	General purpose I/O
13	IO2	IO2	bidirectional	General purpose I/O
14	IO1	IO1	bidirectional	General purpose I/O
15	IO4	IO4	bidirectional	General purpose I/O
16	IO5	IO5	bidirectional	General purpose I/O
17	IO6	IO6	bidirectional	General purpose I/O
18	IO7	IO7	bidirectional	General purpose I/O
19	IO15	IO15	bidirectional	General purpose I/O
20	IO16	IO16	bidirectional	General purpose I/O
21	IO17	IO17	bidirectional	General purpose I/O
22	IO18	IO18	bidirectional	General purpose I/O
23	IO8	IO8	bidirectional	General purpose I/O
24	IO9	IO9	bidirectional	General purpose I/O
25	IO10	IO10	bidirectional	General purpose I/O
26	3V3 OUT	N/A	power output	Switchable 3.3VDC output (see 5.1.2)
27	GND	GND	power ground	GND connection
28	VIN	N/A	power input	DC Power input port (see 5.1.1)

Table 1: Walter pin definitions

4.1.2 Internal Pins

Table 2 contains the pin descriptions of the internally connected GPIO pins on Walter. These are necessary for either communication between components on the board or reserved for other purposes and thus not available for external use.

ESP pin	Description
IO19	USB D-
IO20	USB D+
IO46	LTE_WAKE0
IO48	LTE_UART0_TX (See 4.2)
IO14	LTE_UART0_RX (See 4.2)
IO21	LTE_UART0_RTS (See 4.2)
IO47	LTE_UART0_CTS (See 4.2)
IO45	LTE_RESET

Table 2: Walter Internal Pin Definitions

4.1.3 Testpoints

Walter contains 22 testpoints on the bottom of the board that serve multiple purposes. You can use these pins for debugging, interfacing and/or flashing of the Sequans GM02SP and the ESP32-S3-WROOM module.

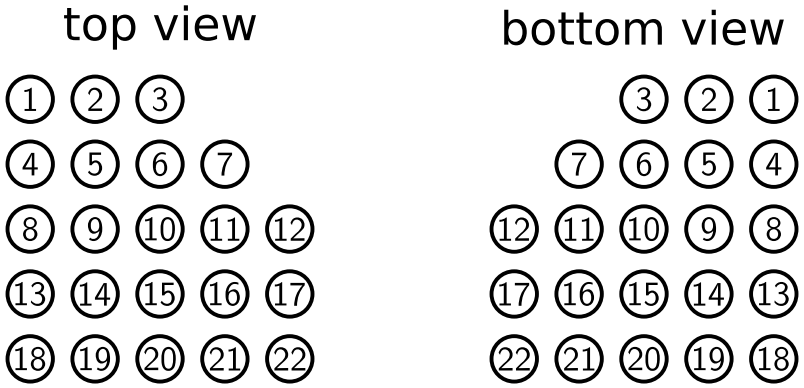


Figure 2: Walter testpoints

Number	Description
1	Sequans GM02SP JTAG TDO
2	Sequans GM02SP JTAG TCK
3	Sequans GM02SP JTAG TRSTN
4	Sequans GM02SP JTAG TMS
5	Sequans GM02SP JTAG TDI
6	Sequans GM02SP PS status
7	Sequans GM02SP RES/FFF_FFH
8	Sequans GM02SP RX0
9	Sequans GM02SP TX0
10	Sequans GM02SP TX1
11	Sequans GM02SP RX1
12	Sequans GM02SP RX2
13	Sequans GM02SP CTS0
14	Sequans GM02SP RTS0
15	Sequans GM02SP CTS1
16	Sequans GM02SP RTS1
17	Sequans GM02SP TX2
18	Walter input power
19	3V3 output (not switched)
20	Sequans GM02SP 1V8 output
21	Ground
22	ESP32-S3 GPIO3 (strapping pin)

4.1.4 Others

Not all pins of the ESP32-S3 and Sequans GM02SP on Walter are internally connected, available through physical pins or testpoints. These pins are either reserved for use by the component itself or deemed not necessary to be available externally.

4.2 Sequans GM02SP UARTs

The Sequans GM02SP module has 3 hardware UART interfaces. Only UART0 is connected to the ESP32-S3 Wroom Module on Walter as shown in Table 2. Communication between modules is possible with AT-commands. Please refer to the corresponding AT command reference manual of the Sequans GM02SP for all possible AT-commands. The UARTs have the following functionality by default:

- UART0 (115200@8N1 with HW handshaking): used for AT commands.
- UART1 (921600@8N1 with HW handshaking): used for manual firmware updates and/or custom software installation.
- UART2 (115200@8N1 no HW handshaking): console log output.

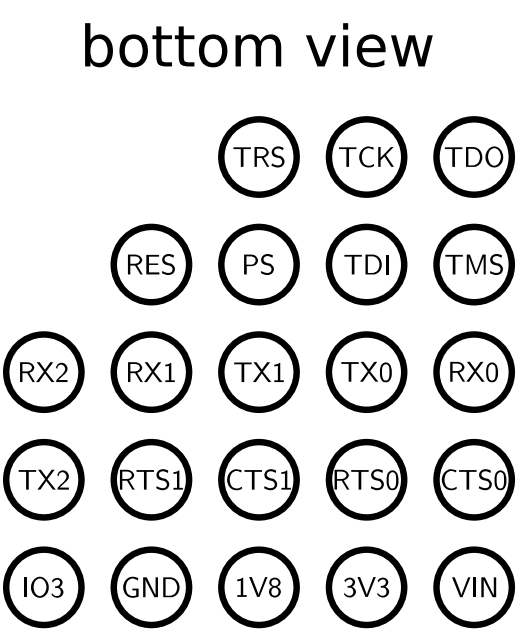


Figure 3: Bottom view of the Walter testpoints with their connection names.

Please note that any UART host should be connected as follows:

- `RX <-> RX`
- `TX <-> TX`
- `RTS <-> RTS`
- `CTS <-> CTS`

5 Electrical and RF Characteristics

5.1 Power

5.1.1 Power Input

Walter can be powered either by connecting a USB-C cable or via the VIN pin (see pinout 4.1.1). VIN accepts 3.0V to 5.5V. Everything on Walter works down to 3.0V, but radio performance is best between 3.3V and 5.5V, so we recommend that range for your design. Below 3.3V the DC-DC converter runs at 100% duty cycle and the internal 3.3V rail follows VIN. The default brownout level of the ESP32-S3 is 2.44V.

The Sequans GM02SP modem is powered directly from VIN. Only the ESP32-S3 and the 3V3 OUT pin are powered through the DC-DC converter (see section 5.1.2). This is why the peak input current rises when VIN drops (see section 5.1.3).

Protection Walter has no overvoltage, surge or reverse polarity protection on VIN. The maximum VIN of 5.5V is also the absolute maximum rating of the modem supply, so VIN must never exceed 5.5V, including spikes caused by hot plugging, long supply cables or inductive loads. A standard TVS diode for 5V lines clamps well above 5.5V and is not sufficient on its own. When VIN comes from an unregulated or external source, place a regulator or an overvoltage protection switch in front of Walter. Add reverse polarity protection when the supply can be connected the wrong way round.

Bulk capacitance Place at least 2 $\times$ 10µF 0603 ceramic capacitors (X5R or X7R) as close as possible to the VIN pin on the carrier board. They supply the current peaks during LTE transmission.

USB-C and VIN To minimize losses, the VBUS line of the USB-C connector and the VIN pin are directly connected to each other on Walter, without a diode. A voltage on one of them is therefore always present on the other:

- When Walter is powered through USB-C, the 5V of the USB port is present on the VIN pin and feeds into the carrier board.
- When Walter is powered through VIN, the carrier supply is present on the USB-C connector and feeds into the USB port of the connected computer.

DO NOT connect a regular USB-C cable while the carrier board powers Walter through VIN, unless the carrier board is designed for it as described below. This can lead to seriously damaging the board, the computer and external peripherals connected to it! For occasional debugging, a USB-C cable with the power lead cut is the simplest solution.

To use a regular USB-C cable while Walter is powered from the carrier board, place an OR-ing element between the carrier supply and the VIN pin, as shown in Figure 4. Whichever source has the highest voltage then powers Walter, and no current can flow from the USB port back into the carrier supply.

- **Schottky diode:** the simplest solution, suited for a 5V carrier supply. Choose a part rated for at least 2A with a low forward voltage. VIN then sits a few hundred millivolts below the carrier supply, so a connected USB port takes over automatically.
- **Ideal diode:** an ideal diode IC, or an ideal diode controller with an external MOSFET. Use this when the carrier supply is low, for example a single Li-ion cell or a 3.3V to 4.2V rail, where the drop of a Schottky diode would push VIN below the recommended 3.3V. It must block reverse current and be rated for at least 1.5A peak. A single MOSFET used as reverse polarity protection is not sufficient, because it conducts in both directions once it is switched on.

The OR-ing element only protects the carrier supply. To also avoid feeding current into the computer, the voltage the carrier delivers on VIN must stay below the USB voltage (nominally 5V). A 5V supply with a Schottky diode, or a supply below 4.5V with an ideal diode, meets this. Do not connect other circuits of the carrier board to the VIN side of the OR-ing element, unless they may be powered from the USB port.

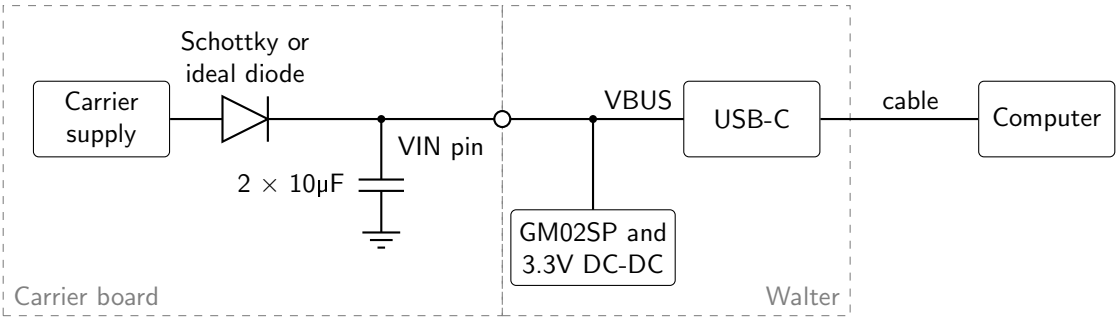


Figure 4: OR-ing element on the carrier board, required when a regular USB-C cable is connected while the carrier board powers Walter.

5.1.2 Power Output

Walter contains a DC-DC converter which takes power from VIN and converts it to a regulated 3.3VDC supply. This rail powers the ESP32-S3 module and the 3V3 OUT pin. Below a VIN of 3.3V the converter runs at 100% duty cycle and the 3.3V rail follows VIN.

The 3V3 OUT pin is switched by a P-channel MOSFET which is controlled by IO0 (pin 4):

- The enable is active low: IO0 low switches 3V3 OUT on, IO0 high switches it off. IO0 is a strapping pin which is pulled high during reset, so 3V3 OUT is off while the ESP32-S3 boots. Pulling IO0 low externally during reset puts the ESP32-S3 in download (DFU) mode.

- The maximum load on 3V3 OUT is 250mA. The 3.3V rail is shared with the ESP32-S3, which needs the remaining capacity of the converter for WiFi and BLE transmission.
- The maximum capacitive load on 3V3 OUT is 1µF. The switch has no slew rate control, so the load capacitance is charged from the 3.3V rail of the ESP32-S3 the moment 3V3 OUT is switched on. A larger capacitance causes a voltage dip that can reset the ESP32-S3.
- 3V3 OUT has no current limit or short-circuit protection. A short or an overload pulls down the supply of the ESP32-S3.

Loads that need more capacitance, 10µF maximum, should use a load switch with slew rate control on the carrier board, or be powered from VIN. A dedicated point-of-load converter powered from VIN is typically an even better choice, and is required for loads above 250mA.

5.1.3 Power Consumption

Table 3 lists the typical current consumption of Walter, measured at VIN = 5.0V.

Operating mode	**Typical current @ VIN = 5.0V**
Deep sleep, modem in PSM	9.5µA
Modem in PSM, CPU running at 10MHz	10mA
Modem in PSM, CPU running at 240MHz	60mA
LTE transmit peak, CPU running at 240MHz	450mA

Table 3: Typical current consumption

For the current consumption of the individual radio modes, please refer to the datasheets of the ESP32-S3-WROOM-1 (WiFi and BLE) and the Sequans GM02SP (LTE-M, NB-IoT and GNSS).

Because the modem is powered directly from VIN, its peak current rises when VIN drops. The worst-case peak input current, with LTE transmitting at 23dBm, WiFi transmitting and 250mA drawn from 3V3 OUT at the same time, is 1.5A at VIN = 3.0V and 1.0A at VIN = 5.0V, including a 20% margin. Size the power supply for these peaks and place the bulk capacitors described in section 5.1.1.

5.2 GPIO

All GPIO pins exposed via the physical pin headers on Walter use 3.3V logic levels and are not 5V tolerant. If you want to connect 5V or other forms of logic, please use a suitable logic level converter or voltage divider.

Please reference the corresponding datasheets for all minimum, maximum and typical ratings of I/O pins of the ESP32-S3-WROOM or Sequans GM02SP Modules that may or may not be exposed on the Walter Development Board.

6 Mechanical information

The drawings in this section show hardware revision v1.6. All dimensions are in mm. Unless stated otherwise, the general tolerance is ±0.2mm. The PCB is 1.6mm thick (±10%). The positions in Table 4 are measured from the centre line of the board (X, positive towards pins 15 to 28 in the top view) and from the USB-C end of the board (Y).

6.1 Top and bottom view

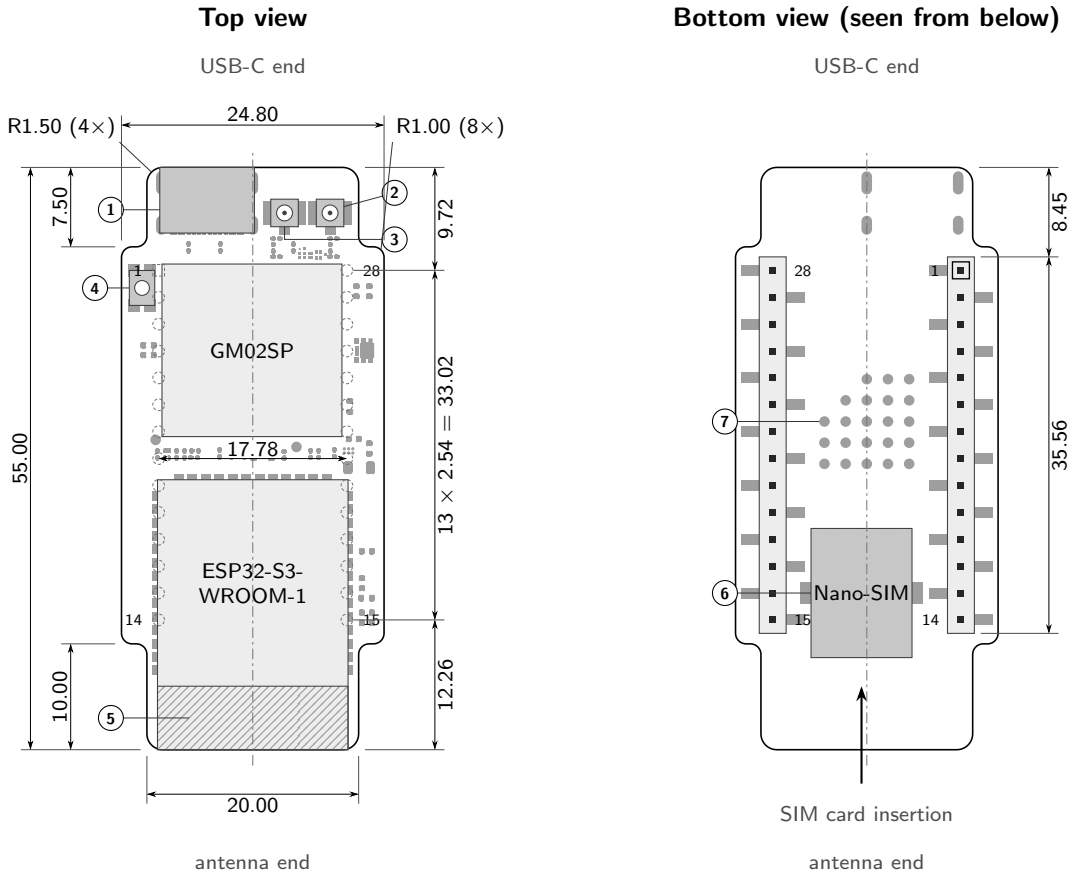


Figure 5: Top and bottom view of Walter. In the top view the header pins are shown with dashed lines, pin 1 is marked with a square. Unit: mm.

Ref.	Feature	Side	X (mm)	Y (mm)
1	USB-C receptacle, centre line	top	−4.31	board edge
2	u.FL connector GNSS antenna	top	+7.30	4.30
3	u.FL connector LTE antenna	top	+3.00	4.30
4	Reset button	top	−10.46	11.38
5	PCB antenna area (P000000100/118)	top	−9.00 to +9.00	49.00 to 55.00
6	Nano-SIM holder, centre	bottom	+0.50	40.20
7	Test pads (section 4.1.3)	bottom	−4.50 to +4.50	19.50 to 28.50
	Pin 1 / pin 28		−8.89 / +8.89	9.72
	ESP32-S3 module body	top	−9.00 to +9.00	29.50 to 55.00
	Sequans GM02SP body	top	−8.58 to +8.42	9.12 to 25.42

Table 4: Position of connectors and components, numbers refer to Figure 5

6.2 Side view and mounting heights

Walter has two 14-pin 2.54mm SMT pin headers on the bottom side (Würth Elektronik 61001418221 or equivalent), with 6.0mm long pins of 0.64mm square. Walter can be soldered into the carrier board, using reflow with pin-in-paste or wave soldering (see section 9), or it can be plugged into two 14-pin sockets. The recommended socket is the Würth Elektronik 61301411821, which is 8.5mm high.

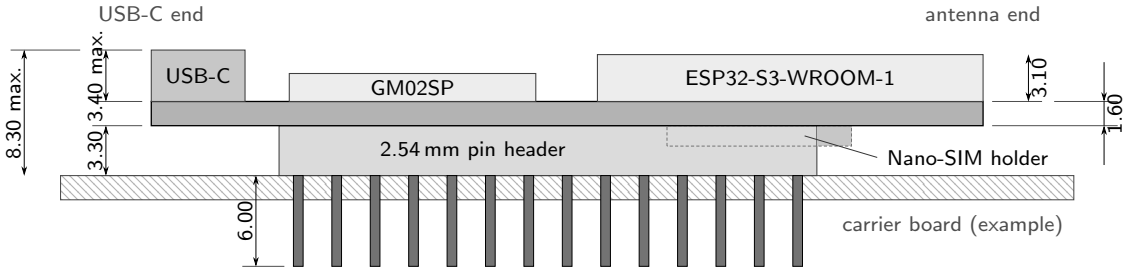


Figure 6: Side view of Walter soldered on a carrier board, seen from the side of pins 1 to 14. Unit: mm.

Parameter (mm)	Soldered	In socket 61301411821
Bottom of the Walter PCB above the carrier board	3.3	11.8
Top of Walter above the carrier board, max.	8.3	16.8
Pin length below the header body	6.0	6.0

Table 5: Mounting heights

The highest parts on the top side are the USB-C receptacle (max. 3.4mm) and the ESP32-S3 module (3.1mm). On the bottom side there are only the pin headers, the test pads and the Nano-SIM holder (1.35mm). When Walter is soldered, there is 3.3mm between the carrier board and the bottom of Walter, and 1.95mm below the Nano-SIM holder. Parts on the carrier board below Walter must stay below these heights.

The Nano-SIM holder is a push-pull type. The card is inserted from the antenna end of Walter, as shown by the arrow in Figure 5. If the SIM card has to be inserted or removed while Walter is mounted, keep the area in front of the holder free. Otherwise, insert the SIM card before mounting Walter.

6.3 Recommended carrier board footprint

Figure 7 shows the recommended footprint, seen from the top of the carrier board with Walter mounted on top. The same footprint is used for soldering Walter and for the sockets: 28 plated through holes of 1.02mm with 2.0mm pads, pin 1 at the USB-C end. The hatched area is the WiFi/BLE antenna keepout described in section 6.5. KiCad footprints for both mounting options are available at https://github.com/QuickSpot/walter-hardware/tree/main/walter/footprint.

6.4 u.FL WiFi/BLE version (P000000119)

P000000119 uses the ESP32-S3-WROOM-1U module, which is 19.2mm long instead of 25.5mm and has a u.FL connector for an external WiFi/BLE antenna. The board outline, the pinout, the other connectors and all other dimensions are the same as for the versions with a PCB antenna. The last 6.3mm at the antenna end has no parts, and no antenna keepout is needed on the carrier board. The module is 3.2mm high. The position of the u.FL connector on the module is given in the ESP32-S3-WROOM-1U datasheet.

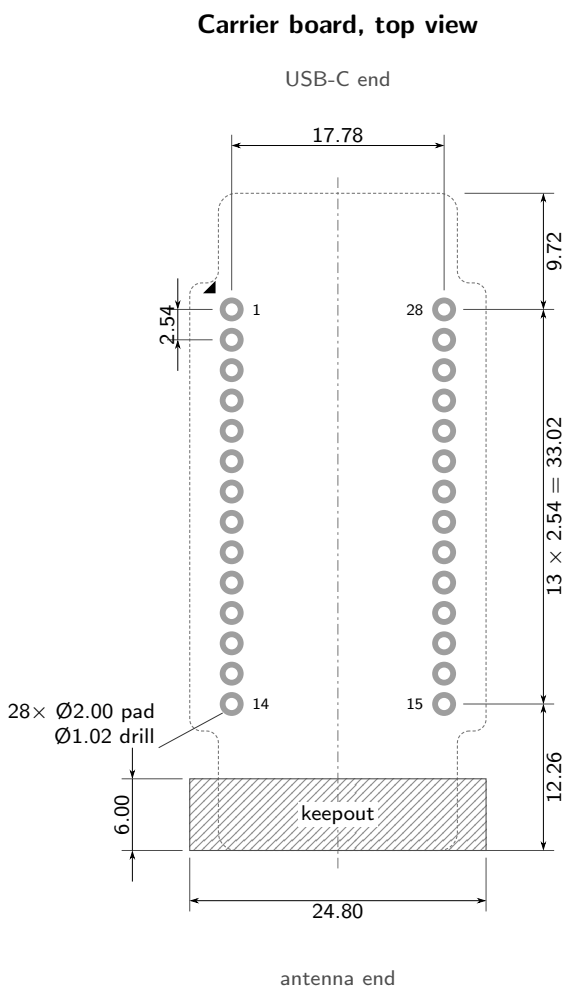


Figure 7: Recommended carrier board footprint. Unit: mm.

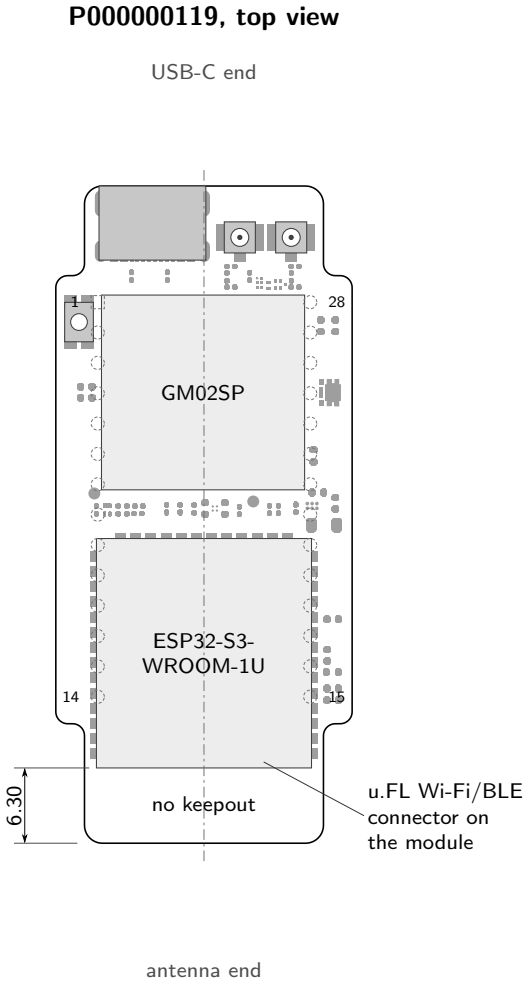


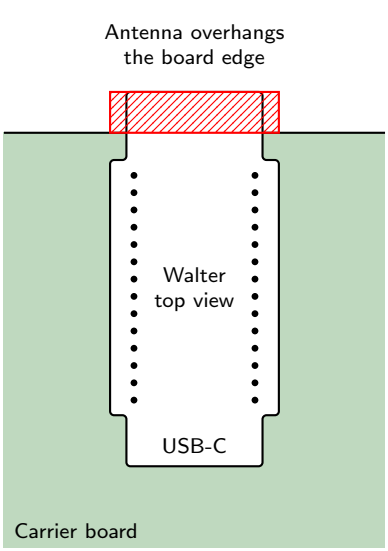
Figure 8: Top view of P000000119 (u.FL WiFi/BLE version). Unit: mm.

6.5 WiFi/BLE antenna keepout

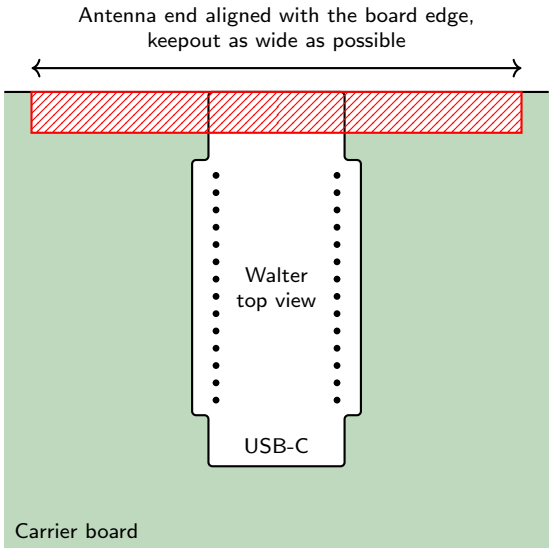
The PCB antenna of the ESP32-S3 module is located at the end of Walter opposite to the USB-C connector. Copper and metal close to the antenna detune it and reduce the WiFi and BLE range, so the carrier board must respect the following guidelines. They apply to the versions with a PCB antenna (P000000100 and P000000118). P000000119 has a u.FL connector for an external antenna and needs no keepout.

- **Preferred:** place Walter at the edge of the carrier board, with the antenna end overhanging the edge or with the carrier board cut away below and next to the antenna.
- **Keepout zone:** keep an area of at least 24.8mm (the full width of Walter) by 6mm, measured from the antenna end of Walter, free of copper on all layers. No traces, ground pour, vias or components are allowed in this zone.
- **Alternative:** when the antenna cannot overhang, align the antenna end of Walter with the edge of the carrier board. Walter can then be placed anywhere along that edge, also in the middle. Extend the 6mm deep keepout zone along the board edge and make it as wide as possible on both sides of Walter.
- Keep metal parts such as enclosures, batteries, displays, screws and cables at least 15mm away from the antenna in all directions.
- Route USB, UART and switching power supply traces away from the antenna end.

Always verify the WiFi and BLE performance in the final product and enclosure.



(a) Preferred: at the edge of the carrier board



(b) Alternative: antenna end at the board edge

Figure 9: WiFi/BLE antenna keepout (hatched): no copper on any layer. Minimum zone 24.8mm × 6mm from the antenna end of Walter.

7 Software

7.1 Libraries and frameworks from QuickSpot

Every Walter comes pre-flashed with an example application. This demonstration firmware requires the LTE and GNSS antennas to be connected and performs the following tasks:

1. Configure the modem for LTE-M connectivity with the network-pushed APN.
2. Check and update the GNSS assistance data if required.
3. Try to establish a GNSS fix. The firmware attempts this up to 5 times, with a maximum duration of approximately 66 seconds per attempt.
4. Upload the data to the walterdemo.quickspot.io portal.

To use this pre-flashed demonstrator, install the LTE and GNSS antennas and insert a SIM card that supports LTE-M. Make sure the GNSS antenna has a clear view of the sky and allow Walter about 10 minutes to connect and upload its first message. Every Walter identifies itself using the MAC address printed on the bag in which it was shipped.

If you do not want to upload your position to the public walterdemo.quickspot.io platform, or if you want to run your own code, which is what Walter is designed for, you can flash your own firmware. You can easily program and upload your own firmware for Walter using Arduino, ESP-IDF, MicroPython or Zephyr. Please refer to the "Getting Started with Walter" guide available on our GitHub to get you up and running fast. The original demonstration firmware can be restored at any time with the online flasher (see section 7.2).

7.2 Online flasher

Walter can be flashed straight from your browser with the online flasher at https://www.quickspot.io/flash.html, without installing drivers or a toolchain. Connect Walter to your computer with a USB-C cable, close any other program that uses its serial port and use a browser with Web Serial support, such as Chrome or Edge. The online flasher offers:

- Ready-made firmware images, for example to update the firmware of the Sequans GM02SP modem or to restore the original factory firmware.
- Flashing your own firmware: upload a .bin file built with Arduino, ESP-IDF or PlatformIO. Use offset `0x10000` for application images and `0x0` for merged images. Optionally, the entire flash memory can be erased first.
- A serial terminal (9600 to 921600 baud) which opens automatically once flashing is done.

7.3 AT commands

The full set of supported AT commands can be found in the documentation of the Sequans GM02SP. This can be downloaded from this link: https://www.renesas.com/eu/en/document/mah/ryz024-modules-command-users-manual?language=en.

7.4 Manual SFU of the GM02SP

The easiest way to update the firmware of the GM02SP is the online flasher (see section 7.2). The manual procedure below is an alternative which uses the UART of the modem directly.

The second hardware UART of the GM02SP, namely `UART1`, is used for manual firmware upgrades. To do a manual serial SFU of the GM02SP in Walter you must connect `UART1` via the testpads on the bottom of the module. Use an FTDI UART and make the following connections:

- `Walter RX1 <-> FTDI RX`
- `Walter TX1 <-> FTDI TX`
- `Walter RTS1 <-> FTDI RTS`
- `Walter CTS1 <-> FTDI CTS`
- `Walter GND <-> FTDI GND`
- `Walter Vin <-> +5VDC`

Make sure that the 5V power source which is powering Walter and the FTDI share a common ground. To enable smooth updating it is best not to connect the USB-C on Walter to the computer on which the SFU software is running. Now follow the instructions which are bundled with the SFU release.

8 Operating conditions

The module can operate in a wide range of temperatures and conditions. The following are guidelines in which the module is guaranteed to work correctly.

Parameter	Units	Minimum rating	Typical rating	Maxium rating
Working temperature	°C	-40		85
Storage temperature	°C	-40		100
Humidity	%RH	10		90
Storage humidity	%RH	5		90

Please note that no condensation may occur on the PCB and components.

9 Manufacturing information: soldering and MSL

The Walter module can be assembled using either a reflow soldering process with pin-in-paste technology or a wave soldering process.

When the module is subjected to reflow soldering, it shall be handled in accordance with Moisture Sensitivity Level 3 (MSL 3) requirements. MSL handling is not required for wave soldering processes in which the module body temperature remains below 125°C.

For reflow soldering, the profile shown in Figure 10 is recommended. This profile is suitable for use with SAC305 lead-free solder paste.

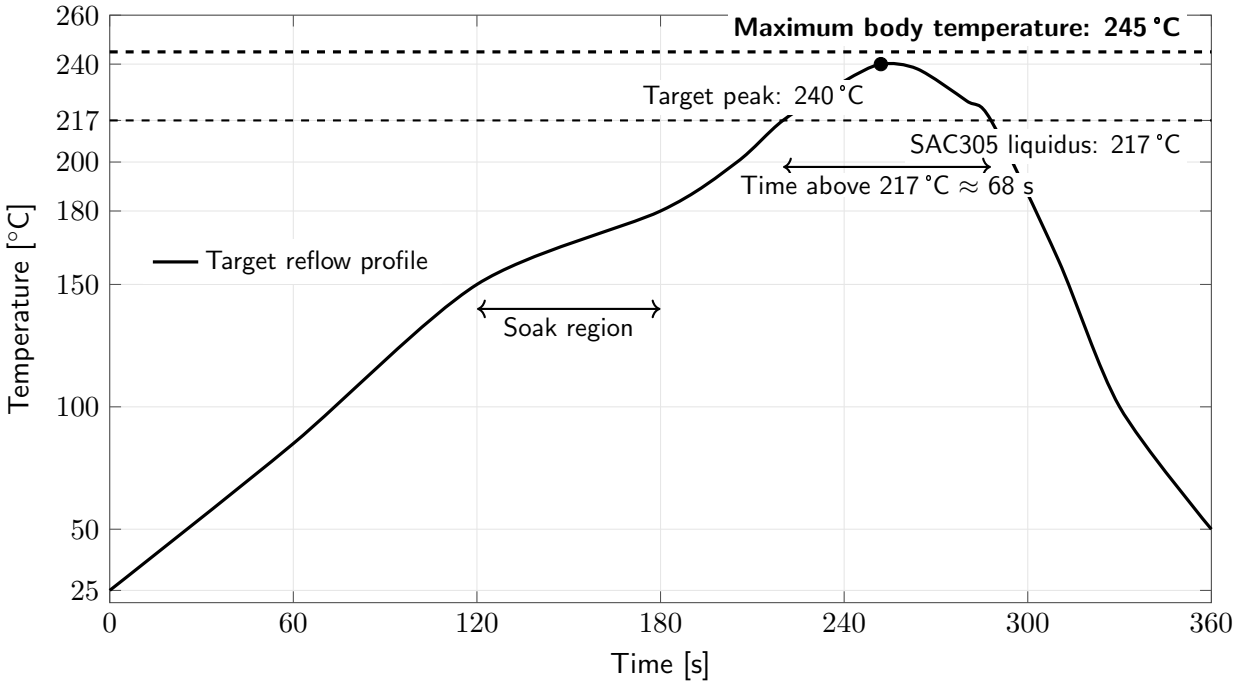


Figure 10: Recommended lead-free reflow profile for the Walter module.

10 Prohibited Use and Resale Statement

This electronics manual serves as a legal document outlining the terms of use and resale of the electronics herein described. By accessing, using, or reselling these electronics, you agree to abide by the terms outlined herein.

10.1 Prohibition on Resale to Third Countries

Users of these electronics are strictly prohibited from reselling them to third countries listed under the Export Control Classification Number (ECCN) or mentioned in Section 2.903 of the Federal Communications Commission (FCC) rules. Any such resale is considered a violation of export control regulations and may result in legal consequences.

10.2 Prohibition on Military Use

These electronics may not be used for military purposes without the prior written consent of the manufacturer. Any unauthorized use for military applications is strictly prohibited and may result in legal action.

10.3 Compliance with Applicable Local Laws

Users of these electronics are obligated to comply with all applicable local laws and regulations. This includes but is not limited to Section 5301 of the Anti-Drug Abuse Act of 1988, 21 U.S.C. 862. Any use of these electronics for purposes contrary to local law is strictly prohibited.

10.4 Manufacturer's Liability

The manufacturer holds no liability for any misuse or unauthorized use of these electronics. Users assume full responsibility for their actions and agree to indemnify and hold harmless the manufacturer from any claims, damages, or liabilities arising from such misuse.

11 Certification

Walter has the following certifications:

- FCC: 2BDGUWALTER
- ISED: Contains 21098-ESPS3WROOM1 and 12732A-GM02SA
- CE: European Economic Area
- UKCA: Great Britain
- RCM: New Zealand and Australia
- Verizon: approved through the Verizon Open Development program (see section 11.9)

11.1 Federal Communication Commission Interference Statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference.
2. This device must accept any interference received, including interference that may cause undesired operation.

CAUTION: Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

NOTE: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

11.2 RF warning statement

To comply with FCC RF exposure compliance requirements, the antennas used for this transmitter must be installed to provide a separation distance of at least 20 cm from all persons and must not be co–located or operating in conjunction with any other antenna or transmitter.

11.3 OEM integrator conditions

This device is intended only for OEM integrators under the following conditions:

1. The antenna must be installed such that 20 cm is maintained between the antenna and users.
2. The transmitter module may not be co–located with any other transmitter or antenna.

As long as the two conditions above are met, further transmitter tests will not be required. However, the OEM integrator is still responsible for testing their end–product for any additional compliance requirements required with this module installed. To ensure compliance with all non–transmitter functions the host manufacturer is responsible for ensuring compliance with the module(s) installed and fully operational. For example, if a host was previously authorized as an unintentional radiator under the Declaration of Conformity procedure without a transmitter certified module and a module is added, the host manufacturer is responsible for ensuring that after the module is installed and operational the host continues to be compliant with the Part 15B unintentional radiator requirements.

The module is limited to OEM installation only. The module is limited to installation in a fixed application. We hereby acknowledge our responsibility to provide guidance to the host manufacturer in the event that they require assistance for ensuring compliance with the Part 15 Subpart B requirements.

IMPORTANT NOTE: In the event that these conditions cannot be met (for example certain laptop configurations or co–location with another transmitter), then the FCC authorization is no longer considered valid and the FCC ID cannot be used on the final product. In these circumstances, the OEM integrator will be responsible for re–evaluating the end product (including the transmitter) and obtaining a separate FCC authorization.

11.4 Antennas

The maximum LTE antenna gain values (accounting for cable attenuation) to comply with the FCC and RED maximum ERP/EIRP limits and with RF Exposure rules:

- LTE band 2 (1900 PCS): 8.0 dBi
- LTE band 4 (AWS-1): 5.0 dBi
- LTE band 5 (850): 9.4 dBi
- LTE band 12 (700a): 8.6 dBi
- LTE band 13 (700c): 9.1 dBi
- LTE band 17 (700b): 8.7 dBi
- LTE band 25 (1900+): 8.0 dBi
- LTE band 66 (AWS-3): 5.0 dBi

An example of an approved LTE antenna is the Taoglas FXUB63.07.0150C.

An example of an approved GNSS antenna is the Taoglas FXP611.07.0092C.

11.5 Limited module procecures

Not applicable

11.6 Trace antenna designs

Not applicable

11.7 End product labelling

This transmitter module is authorized only for use in device where the antenna may be installed such that 20 cm must be maintained between the antenna and users. The final end product must be labelled in a visible area with the following: "Contains FCC ID: 2BDGUWALTER". The grantee's FCC ID can be used only when all FCC compliance requirements are met. The following FCC part 15.19 statement has to also be available on the label:

```
This device complies with Part 15 of FCC rules. Operation is subject to the
following two conditions:
1. this device may not cause harmful interference and
2. this device must accept any interference received,
   including interference that may cause undesired operation.
```

11.8 Manual information to the end user

The OEM integrator has to be aware not to provide information to the end user regarding how to install or remove this RF module in the user's manual of the end product which integrates this module. In the user manual of the end product, the end user has to be informed that the equipment complies with FCC radio–frequency exposure guidelines set forth for an uncontrolled environment. The end user has to also be informed that any changes or modifications not expressly approved by the manufacturer could void the user's authority to operate this equipment. The end user manual shall include all required regulatory information/warning as show in this manual. The maximum operating ambient temperature of the equipment declared by the manufacturer is -40°C to +85°C.

11.9 Verizon

Walter is approved for use on the Verizon network through the Verizon Open Development program and is listed as device 32161 in the Verizon device showcase:
https://opendevelopment.verizonwireless.com/device-showcase/device/32161

The listing contains the following details:

- Network technology: LTE-M (Cat M1)
- Verizon LTE bands: Band 4 (1700MHz), Band 12 (700MHz) and Band 13 (700MHz)
- FoTA client: LWM2M, using the Verizon FoTA solution
- Security level 2

Walter is approved by Verizon as a modem, not as a module. End products that use Walter therefore do not need to go through Verizon certification themselves: design Walter into your product and it can connect directly to the Verizon network. Please contact us if you have questions about using Walter on the Verizon network.

12 Orderable part numbers

P000000100 is the default version of Walter. P000000118 and P000000119 are optional versions which are available through direct ordering from DPTechnics.

Part number	Description	HS	JPHTS	ECCN
P000000100	16MiB Flash, 2MiB PSRAM, PCB antenna	851762	851762090	5A992.C
P000000118	16MiB Flash, 8MiB PSRAM, PCB antenna	851762	851762090	5A992.C
P000000119	16MiB Flash, 8MiB PSRAM, u.FL WiFi/BLE	851762	851762090	5A992.C

13 Legal information

This module is manufactured by DPTechnics bv. We are not responsible for any product this module is part of. This datasheet is made with great care for detail but it is possible the datasheet will be updated with more accurate data in te future. Users of DPTechnics bv products can contact us by letter, telephone or email. For inquiries regarding the permissible use, resale, or any other questions related to these electronics, please contact the manufacturer. By using or reselling these electronics, you acknowledge that you have read and understood the terms outlined in this manual and agree to fully comply with them.

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